WEEK1 -ASSIGNMENT

1. Artificial Intelligence (AI) refers to the ability of a computer system or machine to perform tasks that would normally require human intelligence. These tasks include things like understanding language, recognizing images, making decisions, solving problems, and even learning from experience. In simple terms, AI is about making machines "smart." Instead of just following fixed instructions, an AI system can observe patterns in data, adapt to new situations, and improve its performance over time without being explicitly reprogrammed for every scenario.

2.

Feature	**Artificial Intelligence (AI)**	**Machine Learning (ML)**	**Generative AI (Gen AI)**	**Agentic AI**
In Simple Words	The big umbrella. Making computers smart enough to mimic human behavior.	The math/stats part of AI. Teaching computers to learn from data without hardcoding rules.	The creative AI. It takes a prompt and actually *creates* new text, code, or images.	The independent worker. It doesn't just talk; it thinks, plans, and executes tasks on its own.
How it Works	Uses rules, logic, or data to solve problems and make decisions.	Finds hidden patterns in massive datasets to make predictions.	Trained on huge data to predict the next best word/pixel to make	Uses LLMs as a "brain" to use tools, make a plan, and fix its own mistakes.

Feature	Artificial Intelligence (AI)	Machine Learning (ML)	Generative AI (Gen AI)	Agentic AI
			something new.	
The "Vibe"	"I can play chess and follow complex rules."	"Give me data, and I'll predict if a customer will churn or not."	"Write a 500-word essay about photosynthesis in the style of Shakespeare."	"Book me a flight, find a hotel within my budget, and add it to my calendar."
Real-World Example	Non-playable characters (NPCs) in video games.	Netflix/Spotify recommendation algorithms.	ChatGPT, Midjourney, GitHub Copilot.	An AI assistant that can autonomously debug code, run tests, and deploy a fix.
Human Effort Needed	High if rule-based; relies on programmers setting the boundaries.	Medium. Humans need to clean the data and tweak the models.	Low to Medium. You just give it a prompt, but you might need to iterate.	**Very Low.** You give it a final goal, and it figures out the step-by-step process.

3.

Hugging Face is the leading open-source platform for sharing and using AI/ML models. It hosts over 500,000 models. Here are three notable ones:

1. BERT (Bidirectional Encoder Representations from Transformers) Model ID: google-bert/bert-base-uncased BERT is developed by Google and is one of the

most influential NLP models ever created. It reads text in both directions (left-to-right AND right-to-left) simultaneously, giving it a deep understanding of context. BERT is used for tasks like question answering, sentiment analysis, and text classification. It's pre-trained on Wikipedia and BookCorpus and can be fine-tuned for specific tasks with minimal data.

2. GPT-2 (Generative Pre-trained Transformer 2) Model ID: openai-community/gpt2 GPT-2 is OpenAI's open-source text generation model. It can generate coherent and contextually relevant paragraphs of text given a prompt. It was trained on 40GB of internet text. GPT-2 works as a decoder-only transformer and is commonly used for creative writing, chatbots, and text completion tasks. It laid the foundation for the much larger GPT-3 and GPT-4 models.

3. Stable Diffusion (Stability AI) Model ID: stabilityai/stable-diffusion-2-1 Stable Diffusion is a text-to-image generative model. Given a text prompt like 'a futuristic city at sunset', it generates high-quality images. It works using a technique called Latent Diffusion — it compresses images into a smaller latent space and gradually denoises random noise into a meaningful image. It's widely used for AI art generation, design prototyping, and creative applications.

4.

Feature	**Traditional Programming**	**AI-Assisted Development**
The Core Role	You are a **writer**. You manually type out syntax, loops, and logic.	You are an **editor/reviewer**. You prompt the AI, read its output, and tweak it.
Stuck on a Bug?	You spend 2 hours scrolling through Stack Overflow and documentation.	You paste the error into the AI, and it explains the fix in 5 seconds.

Feature	Traditional Programming	AI-Assisted Development
Boilerplate Code	You have to manually write all the repetitive setup code (like setting up an Express server or database connection).	You type // set up express server and the AI autocompletes the whole thing.
Learning Curve	High. You have to memorize syntax rules and language quirks.	Shorter. You need to understand logic and concepts, but the AI handles the strict syntax.
Brainpower Spent	80% on syntax and debugging, 20% on system architecture.	20% on syntax, 80% on high-level logic and system design.

5. An AI hallucination occurs when a large language model (LLM) confidently generates information that sounds plausible but is factually incorrect, made-up, or completely false. The model doesn't "know" it's wrong — it's producing what statistically seems like a reasonable answer based on its training data, even when that answer doesn't exist in reality.

Why does it happen? LLMs are trained to predict the next most likely token (word) in a sequence. They don't actually "look up" facts — they pattern-match based on training. When asked about something obscure or outside their training data, they sometimes fill in the gap with plausible-sounding but fabricated content.

Ex: Prompt: "List five published research papers by Dr. Rajesh Gupta on quantum cryptography with their DOI numbers." AI Response: The model confidently lists five papers with titles, journal names, years, and DOI numbers. Reality: None of these papers exist. Dr. Rajesh Gupta may be a real person, but the papers, journals, and DOI numbers are entirely fabricated by the model.